

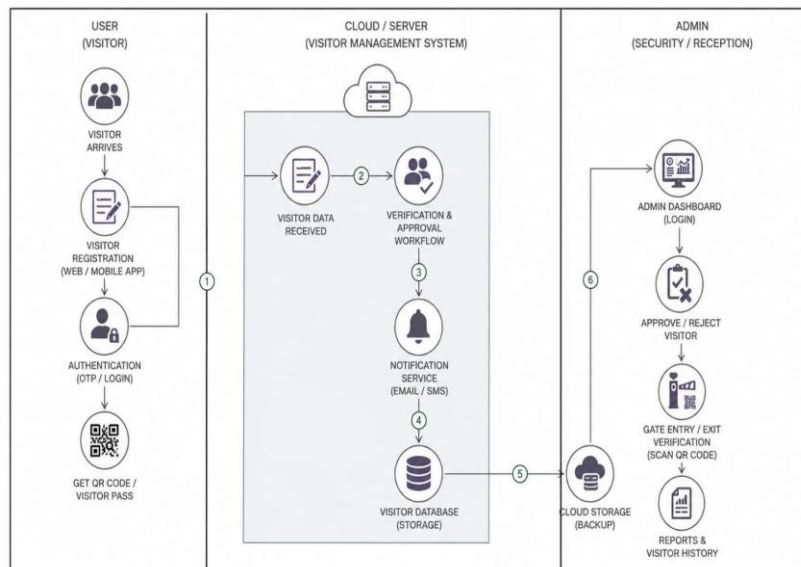
Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	02/07/2026
Team ID	23A51A42B7
Project Name	Visitor Management System
Maximum Marks	4 Marks

Technical Architecture : A layered architecture with UI, Application Server, Database, and Admin Dashboard for secure registration, data storage, and efficient admin control of visitor records.

Example: A visitor registers online, receives a QR code after approval, scans it at the entrance, and the system records the check-in and check-out details securely.



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Visitor Registration , Login, dashboard	HTML, CSS, JavaScript / React Js etc.
2.	Application Logic-1	Visitor Registration & Login	Python (Flask)
3.	Application Logic-2	Visitor Approval Process	Python (Flask)
4.	Application Logic-3	Visitor check in , check out	Python
5.	Database	Store visitor Details , Employee Details , visit History	MySQL
6.	Cloud Database	Cloud based Database	AWS RDS (My SQL)
7.	File Storage	Visitor Photo, ID proof	AWS S3 Storage
8.	External API-1	Email notification	SMTP/ Sendgrid API
9.	External API-2	SMS Notification	Twilio API
10.	Machine Learning Model	Face recognition	Open CV +Face recognition
11.	Infrastructure (Server / Cloud)	Application Hosting	AWS EC2/Local Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend development	React js ,flask ,bootstrap
2.	Security Implementations	Password encryption , Authentication ,Role based access	e.g. SHA-256, JWT authentication ,HTT.
3.	Scalable Architecture	3 – tier architecture	React js ,flask ,My SQL

S.No	Characteristics	Description	Technology
4.	Availability	Cloud hosting with backup	AWS EC2, AWS RDS
5.	Performance	Fast database queries API Optimization , image compression	My SQL indexing , REST API